Scott Moon

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Research Interests

Self-organizing systems, order from chaos, complex behavior following from simple rules

Dynamical systems and stochastic processes

Mathematical biology

Education

West Chester University — *West Chester, PA*
MS in Applied and Computational Mathematics — *Aug 2021 – Dec 2023*

- 4.0 GPA

Penn State — *Erie, PA*
BS in Mathematics — *Aug 2014 – July 2017*

Presentations

"*Mayfly Mathematics: How Mayflies Synchronize*," MAA EPaDel Section Meeting, Media, PA, April 2023

"*Cardiac Action Potential: The Luo-Rudy Model*," WCU Research Competition, West Chester, PA, April 2022

Research Experience

Research Assistant for Dr. Allison Kolpas — *Spring 2022 - Fall 2023*

- Studied dynammical systems representing mayfly emergence times, looking for conditions for synchrony (stable periodic points)
- Developed a model for predicting the final weight of mayflies, used as a proxy for fecundity in population study
- Programmed an agent based model to simulate synchrony and population dynamics

Research Aid for Dr. Frank Fish — *Fall 2022*

- Created a program for measuring turning radius of swimming turtles from video data

Undergraduate research project with Dr. Joseph Previte — *Fall 2016*

- Studied properties of certain matrices arising in finite calculus

Professional Experience

Senior Data Analyst — *2024 – present*
Charles Schwab

- Statistical analysis for classifying customer web and trade behavior

Data Engineer — *2022 – 2023*
Hagerty

- Developed Airflow ETL pipelines in Python, used Snowflake for data warehousing, and used DBT for view creation
- Used Terraform for infrastructure-as-code for AWS resources
- Created custom Python library for automated schema management, metadata, and dynamic SQL queries
- Wrote technical documents for architecture and development proposals

Data Engineer — *2020 – 2022*
Vanguard

- Used PySpark for ETL of unstructured and structured data; running on AWS EMR cluster
- Set up and managed backend for ThoughtSpot BI tool running on AWS EC2

- Implemented infrastructure-as-code using CloudFormation and CDK in Python
- Created data governance system with Boto3, IAM, and Glue Data Catalog
- Created custom data quality system to check completeness, accuracy, and timeliness

Modeling Analyst and Controls Engineer *2017 – 2020*
Enersys

- Designed the ESS (Energy Storage System) architecture and wrote the specifications
- Created an ESS simulation to model cost savings, including a web applet using React and Django
- Implemented the ESS control program in Python running on an embedded Linux system
- Architected PostgreSQL database for storing system data and managing live communication
- Created a Node web application and REST API for monitoring and controlling the ESS
- Led an FMEA and analyzed 6 months of live testing data with R

Grants and Awards

- Viorel Nitica Scholarship, 2022
- 1st place in West Chester Research Competition, 2022
- Undergraduate Research Award, 2016

Extracirricular Activies

Math Club, *President* (2016-2017) - organized faculty talks and fundraising events

International Student Organization, *Program Director* (2016-2017) - organized community building events, celebratory dinners, and fundraising events

Student Government Association, *Budget Committee Member* (2015-2017)

Musician's Club, *Vice President* (2015-2017) - organized student concerts and coordinated musical entertainment at other student events

Math Tutor - drop-in mathematics tutor at PSU (2015-2016) and WCU (2021-2022)

Projects

Turing Morphogenesis

- Exposition on Alan Turing's original 1952 reaction-diffusion paper from a modern perspective, including a bifurcation analysis and 2D numerical simulation using parallelism.

Cardiac Action Potential: The Luo-Rudy Model

- A simulation and analysis of the Luo-Rudy model (a Hodgkin-Huxley type model for heart cells) propagating through a cardiac fiber using the cable equation

Iterated Games and Markov Chains

- An analysis of reactive strategies for the Prisoner's Dilemma using Markov Chains and evolutionary game theory

Walls, an abstract strategy game

- A board game inspired by Theseus and Minotaur, in which both players can move walls and change the maze; wrote a C# implementation and built a physical wooden board and pieces

Technologies

Languages: Python, MATLAB, R, Fortran, JavaScript, C#, C++ C, SQL, HTML, CSS

Frameworks and Platforms: Pyspark, AWS, Hadoop, Airflow, Terraform, REST, Django, GraphQL, React